Native-Resolution Mask R-CNN Ensemble with Flip Test-Time Augmentation

A Solution to the Solar Filament Segmentation Challenge 2026

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Abstract

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. Our pipeline uses two Mask R-CNN models (ResNeXt-101 and ResNet-50 FPN backbones) that work on the full 2048×2048 images without tiling, with anchors shaped for thin, elongated filaments. Each model predicts four flipped views of every image; the views are merged per model and the two models are then merged by mask intersection, followed by simple clean-up rules that target the fragmentation and false-alarm penalties of panoptic quality (PQ). Every annotator's labels are used as separate training samples. On a held-out validation split built from 27-day time blocks, the final pipeline reaches PQ 0.448 (95% CI 0.425–0.470) and a mean Dice of 0.652, against an inter-annotator agreement of PQ 0.336; it scores 0.39 on the public leaderboard. Code, weights and a one-click notebook are public.

1 Introduction

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1]. The task is instance segmentation: every filament must be a separate mask. Submissions are scored with panoptic quality,

$$\mathrm{PQ} = \underbrace{\frac{\sum_{(p,g)\in TP} \mathrm{IoU}(p,g)}{|TP|}}_{\mathrm{SQ}} \times \underbrace{\frac{|TP|}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}}_{\mathrm{RQ}}, \qquad (1)$$

where a prediction and a ground-truth mask form a true positive (TP) only if their IoU is above 0.5. Two properties of the data make this hard. First, filaments are long and thin, so a prediction that breaks one filament into pieces gains one TP but adds false positives (FP). Second, the labels are noisy: when two annotators outline the same image, their masks agree only at PQ $\approx$ 0.336.

Our solution rests on three ideas, each kept only if it beat the validation noise of about 0.010 PQ (Sec. 3): (i) train and predict at the native 2048×2048 resolution, so thin filaments are not lost by down-scaling; (ii) combine two different Mask R-CNN models and keep only the pixels both agree on; (iii) predict each image in four flipped views and average them, which lowers the scores of chance detections. Sec. 2 describes the data, the models and the pipeline; Sec. 3 reports the results, what did not work, and the remaining errors.

[1] www.esairlab.com/

Kaggle Competition, kaggle.com/competitions/filament-segmentation-2026 2026.

2 Methodology

2.1 Data and validation split

We use only the competition data: full-disk GONG Hα images from the MAGFiLO 1.0 dataset [1], 707 annotated training files and 180 unannotated test files, all 2048×2048 JPEGs. The training annotations hold 8,199 filament outlines, with areas from 9 to 37,739 px^2 (mean $\approx$2,120 px^2).

Several annotators per image. The annotation file has 1,154 image entries for the 707 files: 296 files were outlined independently by two or three annotators. The annotators often disagree. Scored against each other with the competition metric, two annotators of the same image reach only PQ $\approx$ 0.336 (SQ 0.63, RQ 0.53), a rough human ceiling for the task. We never merge the annotation sets. For training, every annotator set is its own sample (921 samples from 566 training files); for evaluation, all sets of a file are used, as in the official scorer.

Validation split. We hold out 141 of the 707 files (20%) as a validation set. It was frozen once (seed 42) and used for every model choice. Images taken a few days apart show the same filaments, so a random split would leak near-copies into validation. We therefore group the files into 27-day blocks (about one solar rotation) and keep each block entirely in training or in validation (stratified group 5-fold, strata = number of filaments per file; mean 7.46 in training vs. 7.31 in validation). The test set has no ground truth and was used only for leaderboard submissions. All validation numbers in this report are the official PQ on these 141 files; the public leaderboard (LB) uses about half of the test images.

2.2 Models and training

We use two Mask R-CNN instance-segmentation models from Detectron2, both pretrained on COCO: one with a ResNeXt-101 (32×8d) FPN backbone and one with a ResNet-50 FPN backbone. Mask R-CNN first proposes boxes around possible filaments, then draws a mask inside each box and gives it a confidence score. We use a single class ("filament"). Three changes adapt the models to filaments: (i) small anchor boxes of 8–128 px with aspect ratios 0.2–5, because filaments are long and thin; (ii) for the X101 model,

a finer mask head that draws each mask on a 56×56 grid instead of the default 28×28, which keeps thin filament edges; (iii) both models train and predict on the full 2048×2048 image, without tiling or resizing, so thin filaments keep their fine detail. No pre-processing is applied: solar-disk masking and CLAHE contrast enhancement gave no gain (Sec. 3).

Both models use the same training recipe: every annotator set of the 566 training files is its own sample (921 samples); 6,000 iterations with batch size 2; SGD with learning rate 0.0025, a 200-iteration linear warm-up and a 10× drop at 60% and 85% of training; random horizontal and vertical flips and brightness and contrast changes of ±20%; seed 42. Training ran on one NVIDIA A100 GPU (Google Colab).

2.3 Inference, fusion and post-processing

The full pipeline is shown in Fig. 1. For each test image, each model predicts four views: the original image and its horizontal, vertical and double flip. The predicted masks are flipped back, and only masks with a score of at least 0.2 are kept at this stage. The predictions are then merged in two steps:

(1) **Views (per model).** Masks from different views with IoU ≥ 0.5 are the same filament. The merged mask is their union and its score is the mean over the four views; a view that missed the filament counts as 0. A filament seen in only one view therefore gets a low score.
(2) **Models.** The two models are merged the same way, with IoU ≥ 0.3; the merged mask is the intersection of the two masks and its score is the mean of the two models.

Finally, four post-processing steps target the metric's failure modes: (i) morphological closing with a 5-px elliptical kernel reconnects small breaks inside a filament, which would otherwise count as extra false positives; (ii) masks with a score below 0.6 are removed; (iii) masks smaller than 50 px^2 are removed as noise; (iv) overlapping masks are resolved: the higher-scored mask keeps the shared pixels, and a mask that loses more than 20% of its area is dropped. The remaining masks are written as COCO run-length encodings in the submission file. Fig. 2 shows the output on three validation images.

2.4 Implementation and reproducibility

All code, the frozen validation split and the trained weights of both models are in our public repository[2]; nothing else has to be downloaded. The weights are stored in 94-MB parts with SHA-256 checksums and joined by one command. The notebook `full_pipeline.ipynb` (folder `notebooks/`) runs the whole pipeline on a Colab GPU ("Run all"): it predicts the 180 test images, writes and validates the submission file and scores the validation split. A run from a fresh clone gave the same validation result (PQ 0.4480, TP/FP/FN 1083/495/595) and found 1,255 of the 1,256 submitted test masks again. On an NVIDIA L4 the whole pipeline takes 6.0 s per image (median; 8.2 s at the 95th percentile), 18 minutes for the 180 test images, with 2.9 GB peak GPU memory. The repository also holds 84 automated tests, a README guide for reviewers and a plain-language notebook that explains every step (`notebooks/pipeline_story.ipynb`).

[2]https://github.com/amiruliman592/Kaggle-Competition-

Table 1: From baseline to final pipeline. Validation: official PQ and mean Dice on the frozen 141-file split. LB: public leaderboard, about 50% of the test set.

Step	Val PQ	Dice	LB
Mask R-CNN R50, 1024 px (baseline)	0.391	0.609	0.34
+ all annotator sets as training data	0.398	0.643	0.35
+ native 2048 px, R50	0.423	0.648	0.36
X101 instead of R50	0.426	0.642	0.37
+ ensemble X101 + R50	0.437	0.651	0.38
+ 56×56 mask head, 4 flip views, two-stage fusion (**final**)	**0.448**	**0.652**	**0.39**

3 Evaluation

3.1 Evaluation protocol

All choices were made on a frozen validation split of 141 of the 707 training files (27-day time blocks, so no validation day is also a training day). We score with the official Kaggle PQ: every annotator set of a file is its own ground-truth entry (233 entries), a prediction matches a ground-truth filament at IoU > 0.5, and TP, FP and FN are pooled over all entries. Our implementation gives the same value as the organisers' notebook on the same files (0.3296 both). Dice is `torchmetrics` `DiceScore` on the binary foreground mask (union of all filaments), averaged over entries; the organisers' exact settings are not published.

The labels are noisy: two annotators of the same image reach only PQ 0.336 against each other, and an exact copy of one annotator scores 0.739 on our validation entries, so the reachable PQ is far below 1. To separate real gains from chance we used a paired bootstrap over the 141 files (2,000 draws): a gain smaller than about 0.010 PQ is not distinguishable from noise, and retraining with another random seed changed PQ by only 0.001. The public leaderboard (about half of the 180 test images, two decimals) has a standard deviation of about 0.014, so we used it only as a check. Validation PQ was 0.05–0.06 above the leaderboard on every upload, but the ranking of the steps in Table 1 was the same on both.

3.2 Results

Table 1 shows how each step changed the score. The largest gains came from native 2048-pixel input (+0.025), from combining two models (+0.010) and from the four flipped views with two-stage fusion (+0.011). The last two are above the noise level: +0.039 [0.027, 0.050] for the ensemble against the 1024-pixel model and +0.012 [0.004, 0.020] for the final pipeline against the plain ensemble (95% bootstrap intervals).

Final pipeline. On validation it reaches PQ 0.448 (95% interval 0.425–0.470), with segmentation quality SQ 0.673 and recognition quality RQ 0.665 (1,083 TP, 495 FP, 595 FN), and mean Dice 0.652 (0.628–0.672). The intervals come from 2,000 bootstrap draws of the 141 files. Fig. 3 shows the Dice and IoU of every pair of a predicted and a ground-truth filament that overlap. The matched pairs are tightly grouped (median Dice 0.81, median IoU 0.67); the long tail below the 0.5 cut-off are filaments we found but outlined too differently from the annotator. **One-to-many and many-to-one.** Of the ground-truth filaments, 20 were split into several predicted pieces (19 into two, one into three); 58 of our 933 masks cover two or more

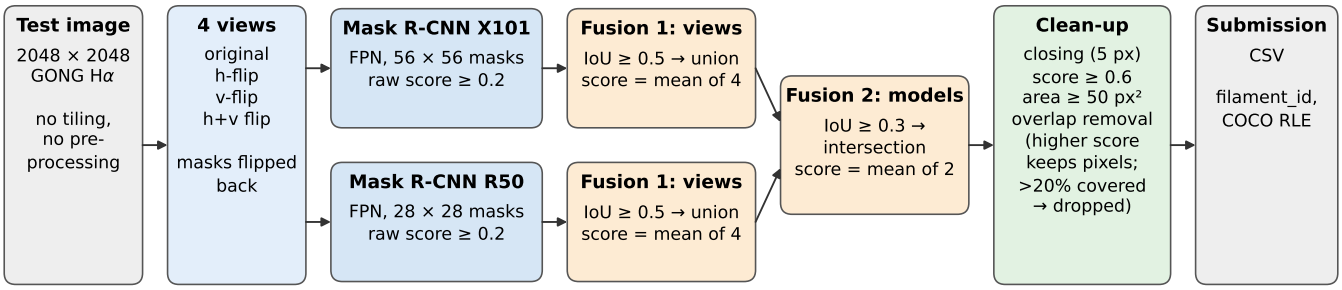


Figure 1: The final pipeline. Each test image is predicted by two Mask R-CNN models on four flipped views; the views of each model are merged first (Fusion 1), then the two models (Fusion 2), and the result is cleaned up before it is written to the submission file.

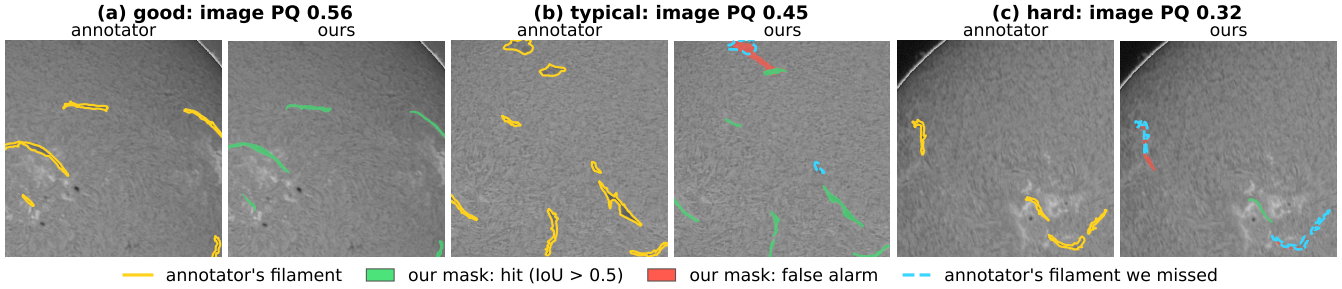


Figure 2: Example segmentations on validation images, chosen by their own PQ: (a) a good image (90th percentile), (b) a typical one (median) and (c) a hard one (10th percentile). For each image, the left crop shows one annotator's filaments and the right crop shows our final masks judged against that annotator. In (c), a mask that covers a real filament but overlaps it with IoU below 0.5 counts as both a false alarm and a miss.

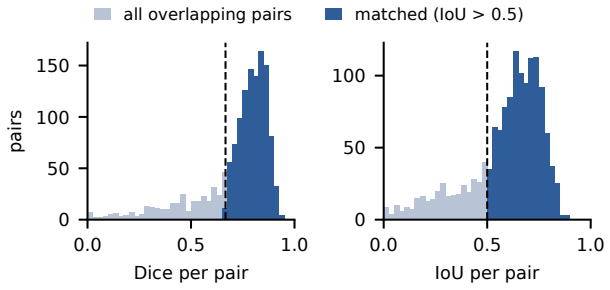


Figure 3: Dice (left) and IoU (right) of every overlapping pair of predicted and ground-truth filaments on validation (all annotator sets). Dark: pairs counted as true positives (IoU > 0.5, dashed line).

ground-truth filaments (55 cover two, 3 cover three). A piece counts when at least half of its area lies inside the other mask. Merging is therefore about three times more common than fragmentation.

Table 2: Rejected ideas (validation official PQ; Δ against the reference of that test).

Idea	ΔPQ	Reason
CLAHE + solar-disk mask	+0.000	no gain
SAM 2 mask refinement	−0.019	tighter than annotators
Mask2Former (Swin-S)	−0.024	twice the merges
3× longer schedule	+0.001	noise, lower Dice
90° rotations + zoom	−0.001	no gain
Copy-paste of filaments	−0.004	more false alarms
Self-training, 2,098 images	−0.002	model too cautious
Third ensemble member	−0.003	same errors as X101
Learned re-scoring	−0.006	too few labels
Retrain on all 707 files	LB =	95% same masks

3.3 What did not work

Table 2 lists ideas that did not pass the 0.010 noise bar. Training-recipe changes were tested on the cheaper 1024-pixel R50 model (reference PQ 0.398), all others on the 2048-pixel models.

SAM 2 draws the dark core of a filament, about 15% smaller than the human outline, which confirms that the labels, not mask sharpness, limit SQ. Self-training used only confident pseudo-labels

(score ≥ 0.85), so faint filaments in the extra images were taught as background.

3.4 Error analysis

We put every validation error of the final pipeline into one cause and computed the PQ if only that cause were fixed. The largest is confidence: 282 ground-truth filaments are found by a raw prediction (IoU > 0.5) but scored below the 0.6 cut-off (+0.072 PQ if kept; 233 of them medium-sized). Next are 305 false alarms that no annotator drew (+0.046), 162 filaments with no prediction near them (+0.043) and 115 near misses with IoU 0.3–0.5 (+0.031). A further 190 "false alarms" were drawn by another annotator of the same image, i.e. label noise rather than model error. By size, medium filaments score PQ 0.430 against 0.493 for large ones; the 12 small filaments ($< 200\,\mathrm{px}^2$) are all missed, which costs at most 0.005 PQ. By station, PQ ranges from 0.418 (El Teide, most false alarms) to 0.468 (Big Bear); with only 20–28 files per station these gaps are within noise.

3.5 Runtime

Inference runs eight forward passes per image (two models × four views) plus fusion and post-processing: median 6.0 s and 95th percentile 8.2 s per 2048×2048 image on one NVIDIA L4 GPU, 18 minutes for all 180 test images, with 2.9 GB peak GPU memory. Training at 2048 pixels needs about 20 GB and was done on an A100.

3.6 Strengths, weaknesses and improvements

Strengths. Every step was chosen on a fixed split against a measured noise level, so the gains carried over to the leaderboard (0.34 to 0.39). The method is simple (two standard Mask R-CNN models, no pre-processing) and a clean clone of the repository reproduces the validation result exactly. **Weaknesses.** The main loss is confidence near the cut-off: many correct medium-sized filaments are scored just below 0.6, and one threshold cannot keep them without adding false alarms. Over-merging of nearby filaments is three times more common than fragmentation, and small filaments are not found. **Improvements.** Better calibrated scores for medium filaments, for example from more labelled data or self-training with a lower pseudo-label threshold, which we did not have time to test; and a rule or model to split masks that join two filaments.

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